

object. In the containment relationship, it contains another object. In a “has-a” relationship, a class becomes an instance variable for the class we are defining.

Inheritance means specialisation. When we inherit from a class, we wish to keep nearly everything in the base class (Superclass). In inheritance, we seek to elaborate on what we receive from the Superclass. We start with the class `Order`. Then, we wish to create a Subclass off of `Order`. Our Subclass is called `RushOrder`.

Class `RushOrder` has everything that `Order` has, but it adds a few instance variables, maybe adds a method or two and overrides a method or two.

The three relationships between classes which form the foundation of Object-Oriented Design are, use, “has-a” and “is-a”

4.2.4 Techniques for using objects

We have spent a lot of time emphasising the difference between a reference and the object to which it refers.

```
JLabel howdy;  
howdy = new JLabel( "How Are Ya?" );
```

Here `howdy` is a reference. We start off by declaring a reference “`howdy`” to an object of type `JLabel`. Then, we instantiate the object by calling its constructor with the `new` keyword, and assign the handle to this instantiation to the reference we declared: “`howdy`”.

```
howdy = new JLabel( "How Are Ya?" );  
Okay, what happens when we execute the following statement?  
JLabel hello; // A new reference  
hello = howdy;
```

Now, both references point to the exact same object. Any changes made from `howdy` will be reflected in `hello`.

4.2.5 Controlling access to methods and variables

`Public` lets clients see the services (methods) the class provides (which means view the interface). The interface is the collective name for all the various methods that are available in the class. Methods should be `public`.

`Private` is the default setting. It hides implementation details. `Private` data members (variables) are only accessible through the public interface (Accessor methods) using public methods. Only utility methods should be made `private`. Utility methods are used only within the class.